



# General stool examination

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## **General Stool Examination**

**STOOL:** The material in a bowel movement. The stool is made up of undigested food, bacteria, mucus, and cells from the lining of the intestines. Also called feces.

**STOOL TEST:** this is a medical diagnostic technique that involves the collection and analysis of the fecal matter. Microbial analysis, microscopy, and chemical tests are among the tests performed on stool samples.

### **The aim of stool examination:**

1. **Look for cause of infection**, such as parasites, bacteria, fungi or virus.
2. **Identify diseases** of the digestive tract.
3. **Check for poor absorption** of nutrients by digestive tract.
4. **Screen for colon cancer** by checking for hidden(occult) blood.

# Characteristics of the stool

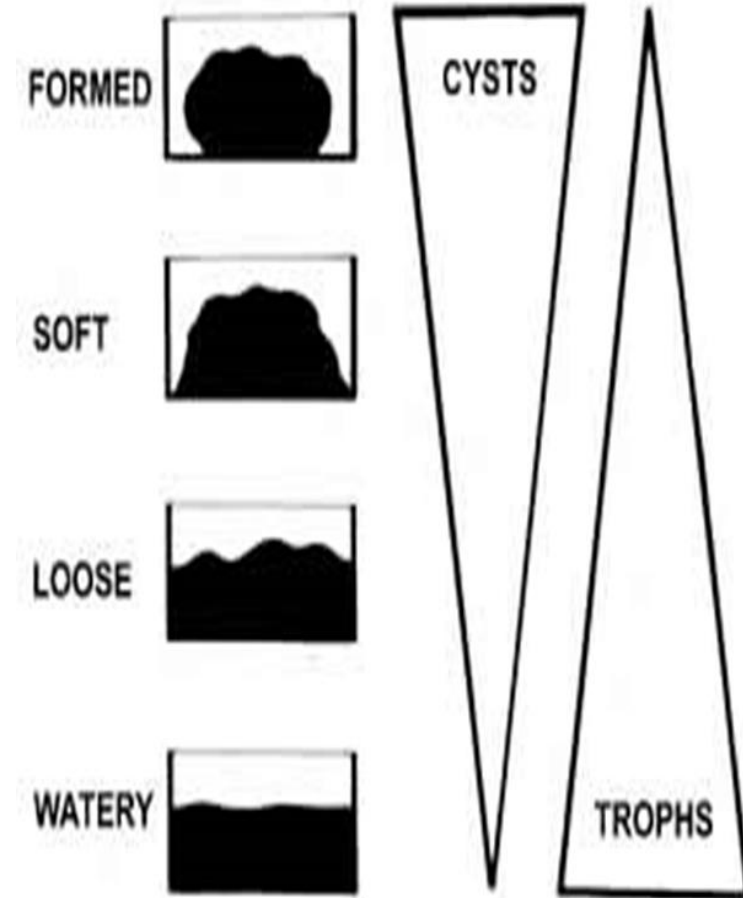
Various points which have to be noted are

## 1) Consistency :

**Formed**, unformed (soft), loose or watery. The **cyst** have been mostly found in the formed stools, while the **trophozoites** have been most abundantly found in watery stools.

## 2) Colour :

- **Normal feces** appear brown and formed or semi-formed.
- **Infant feces** are yellow-green and semi-formed.
- In cases of **malabsorption** of fats the stools are pale bulky and semi solid.
- In **cholera** the stools have a rice water appearance.
- **Person diet** may also lead to alteration in the colour of the stools, spinach green in colour.



### 3. Odor

**Normal:** Aromatic, affected by normal flora and food.

**Abnormal:** foul-smelling stool that can be a sign for a disease e.g infection

### Composition of feces

$\frac{3}{4}$  water,  $\frac{1}{4}$  solid

**Undigested & unabsorbed** food

Bile pigments & salts

**Bacterial & inorganic material (10-20%)** chiefly calcium & phosphates.

**Epithelial cells** that have been shed, **leukocytes** that migrate from blood stream.

**Mucus:** Intestinal secretion include **mucus**

Mucus is present in certain conditions like amoebic or bacillary dysentery.

# Pre specimen processing

- **Patient preparing:**

## Timing

- as soon as possible after onset

## Sample amount

- at least 3 x 5-10 ml fresh stool from patients
- mix with 10% formalin or PVA ( 5%, 10%), 3 parts stool to 1 part preservative
- unpreserved samples for Ag detection and PCR

## Storage

- refrigerate at 4°C; store at -15°C for Ag detection and PCR

## Transport

- 4°C (do not freeze); dry ice for antigen detection and PCR



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## TYPES OF STOOL TEST

- ❖ General stool examination
- ❖ A fecal occult blood test (FOBT) detects the presence of blood in your feces.
- ❖ Stool for Culture and Sensitivity
- ❖ Rotavirus Test. Also called a “nucleic acid detection test” and “isolation in cell culture,” this stool test is used to diagnose a rotavirus infection. ...
- ❖ Giardia Antigen Test. ...
- ❖ Salmonella Culture Test. ...
- ❖ White Blood Cell Test. ...
- ❖ Calprotectin Test.



# Fecal and serological testing in inflammatory bowel disease

- **Fecal calprotectin**
- Neutrophil granulocyte cytosol protein, which acts as a marker of intestinal inflammation.
- +ve correlation with inflammation in Crohn's disease and ulcerative colitis.
- Non-specifically elevated in neoplasia and radiation proctitis too.
- May be used for non-invasive testing to differentiate those who require colonoscopy from those more likely to have functional disease such as IBS.
- Not as effective as calprotectin, as sensitivity for diagnosing pathology is 82% and specificity is 84%.

# Protozoan infections:

- Three species are of particular importance:
  - ❖ *Entamoeba histolytica*
  - ❖ *Giardia intestinalis*
  - ❖ *Cryptosporidium parvum*.
    - All three can give rise to diarrheal illnesses, but the organisms have distinctive features that allow a differential diagnosis to be made quite easily.
    - Other intestinal protozoa of concern, particularly in immunosuppressed patients, include *Cyclospora cayetanensis*, *Isospora belli* and the microsporidia.



# Methods of examination

## A- Macroscopic examination

1. **Consistency, colour & Odor** of stool.
2. The presence of **blood, mucus** or **pus** in the stool is suggestive of amoebic dysentery.
3. The presence of **worms** e.g. *Enterobius vermicularis*, *Ascaris*, tape worm segments, *Taenia* species.

# B- Microscopic examination

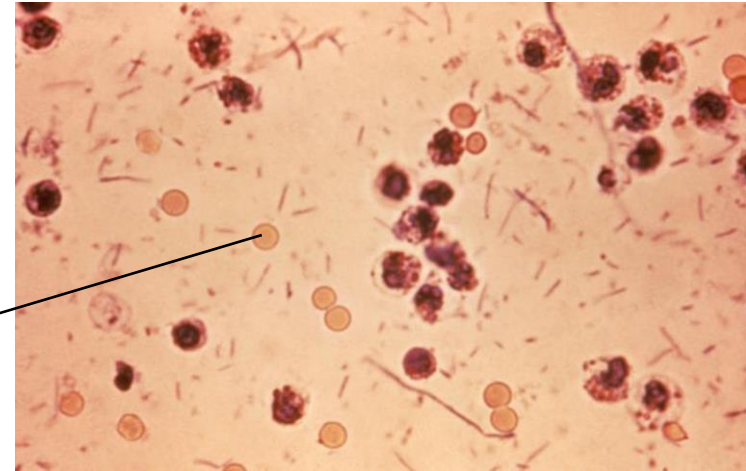
## A-The wet mount :

- ❖ is a preliminary and **direct procedure** that will not only sometimes yield identification of certain parasites, but will enable visualization of WBCs and RBCs.
- ❖ The direct wet mount is the initial and perhaps the most important step and is used as **a screening test** in which trophozoite motility *E. histolytica*, *Giardia lamblia* & *Balantidium coli* can be observed.

## Presence of RBC

- Bright red blood: bleeding in the lower GIT tract.
- Black and tarry blood: bleeding from the upper GIT tract or occult blood.
- Causes of blood in stool:
  - Haemorrhoids
  - GIT tract malignancy
  - Dysentery

R.B.C



Presence of meat fibers and muscle fibers:

- Malabsorption syndrome
- Pancreatic functional defect like cystic fibrosis

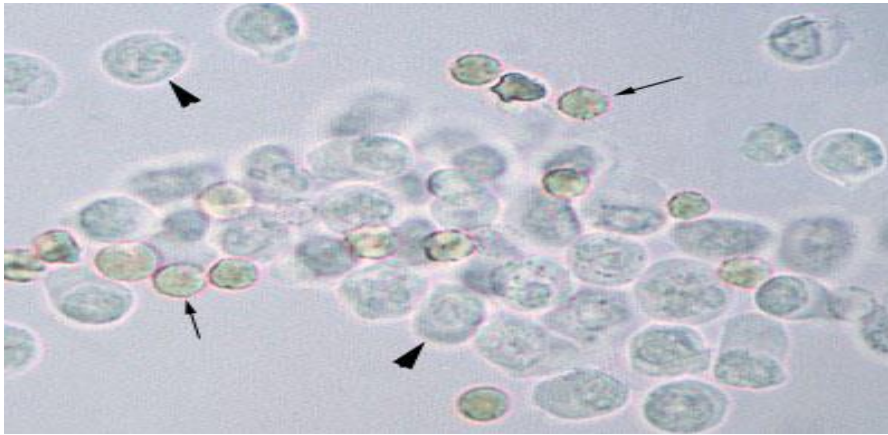


# Microscopic examination: Presence of WBC

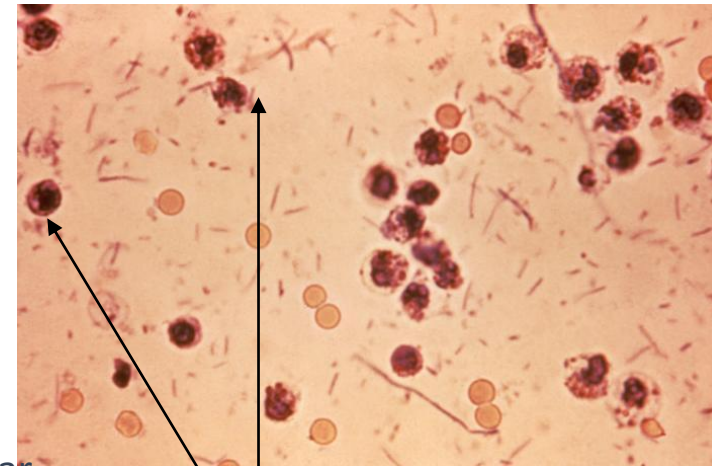
During wet mount preparation to look for WBC's in the smears must be prepared from the area of mucous or watery stool, increased number of WBC's in the stool associated with:

Bacillary dysentery  
Chronic Ulcerative colitis  
Shigellosis  
Salmonella infection

Invasive E. Coli diarrhoea  
Fistula anus or rectum  
Localised abscess  
Amoebiasis & Typhoid



WBC (arrowheads) are larger, colorless and more granular than the redder smaller RBC (arrows), which have no internal texture (although appear slightly biconcave).



WBC in stool (Fecal Leukocytes)  
Normal stool contains (0-1) WBC's

Viral & parasitic infections don't cause WBC in stool



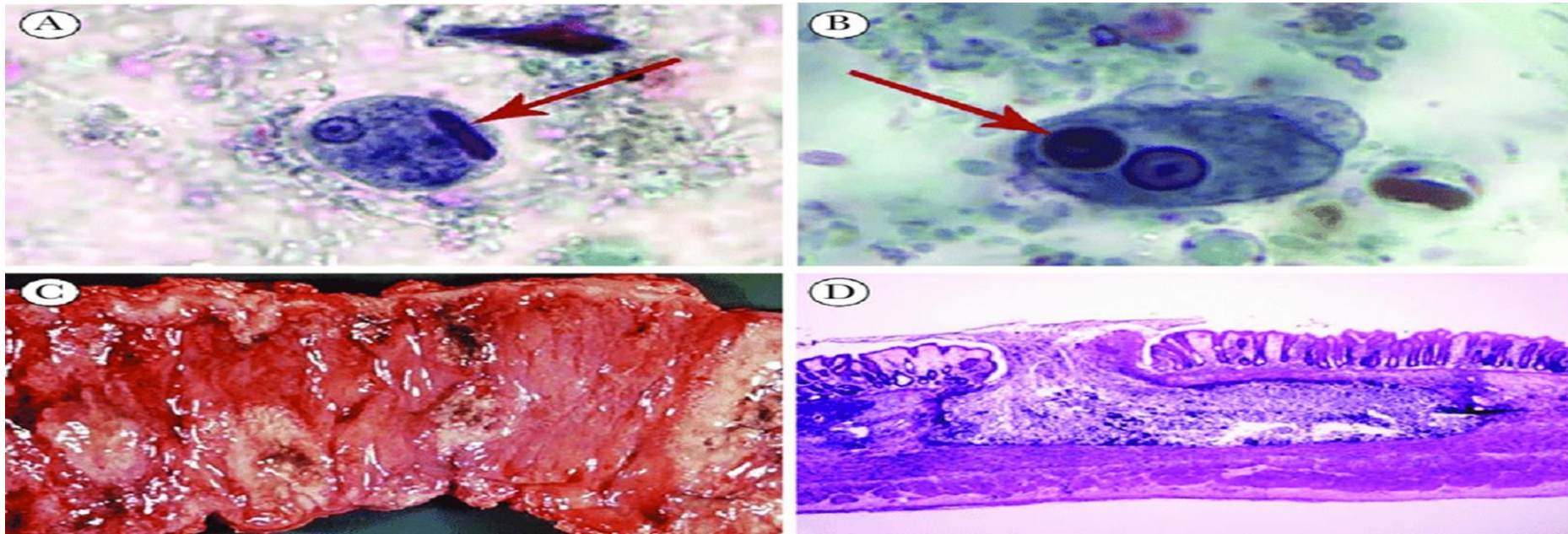
***Entamoeba histolytica*:** In stool and pathological features of intestinal amebiasis.

A, Cyst of *E. histolytica*/*E. dispar* stained with trichrome. Note the chromatoid body with blunt ends (red arrow).

B, Trophozoite of *E. histolytica* with ingested erythrocytes stained with trichrome. The ingested erythrocytes appear as dark inclusions (red arrow). The parasite shows nuclei that have the typical small, centrally located karyosome and thin, uniform peripheral chromatin.

C, Intestinal tissue from a patient with amebic colitis showing multiple ulcers.

D, Classic flask-shaped ulcer of amebiasis (courtesy of the Centers for Disease Control and Prevention).



## 2) Iodine wet mount:

- ❖ Stool is emulsified in a drop of five times diluted solution of Lugol's iodine on a clean glass slide covered with a clean coverslip & examined under microscope.
- ❖ This method is used for the study of **nuclear character of cysts & trophozoites for the identification of the species.**
- ❖ Giardia lamblia cysts iodine stained. Often showing their four nuclei. The fibrils and flagella lie longitudinally along the body (10µm)

